UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. A prototype is an early model of a product.
2. A web browser displays hypertext documents.
3. A higher sampling rate generally produces a more accurate recording.
4. A graphics tablet is used to draw directly into a computer.
5. HTML is used to structure content on web pages.
6. A linear presentation allows users to choose their own path.
7. MPEG is a common family of video compression formats.
8. Streaming requires the whole file to download before playback.
9. Chroma key compositing replaces a colour background with another image.
10. The target audience has no influence on multimedia design.
11. The amplitude of a sound wave determines its loudness.
12. A multimedia project always uses every media element.
13. USB is a plug-and-play connection standard.
14. A scanner is an input device.
15. Multimedia combines text, graphics, audio, video and animation.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A

A. Multimedia	B. Browser	C. Frame rate	D. Storyboard	E. Linear multimedia
F. Vector	G. MIDI	H. Hypermedia	I. Kerning	J. Sans serif
K. Serif	L. Keyframe	M. Sampling	N. Bitmap	O. Scanner

1. converting sound into digital values
2. software that displays web pages
3. font with small strokes at character ends
4. standard for musical notes and instructions
5. integration of text, graphics, audio, video and animation
6. image made of pixels
7. important frame in an animation
8. visual sequence plan
9. presentation without user control of sequence
10. font without small strokes
11. image made of mathematical objects
12. input device that digitises documents
13. frames displayed per second
14. adjustment of space between character pairs
15. linked multimedia elements

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PART C (30 MARKS)

Choose the correct answer for each question.

1. Which is a 3D animation task?
A. Bookmarking
B. Kerning
C. Modelling and rendering
D. Sampling

2. CMYK is used mainly for:
A. Printed output
B. Screen display
C. Sound mixing
D. Animation

3. A browser is used to:
A. Print documents only
B. Edit audio
C. Scan images
D. Display web pages

4. Which image type scales without losing quality?
A. Vector
B. Bitmap
C. Scan
D. Photograph

5. Keyframes are:
A. Important frames in an animation
B. Audio samples
C. Web links
D. Colour profiles

6. Plug and play means:
A. No power needed
B. Manual driver installation only
C. Automatic device recognition
D. Wireless only

7. Tweening creates:
A. Text fonts
B. In-between frames
C. Video codecs
D. Sound effects

8. Cel animation is created by:
A. Printing images
B. Scanning text
C. Recording sound
D. Drawing each frame by hand

9. A presentation where the user controls the sequence is called:
A. Static
B. Linear
C. Printed
D. Non-linear

10. Which is an authoring tool category?
A. RGB
B. Card-based
C. JPEG
D. CMYK

11. Frame rate is:
A. Samples per second
B. Bits per second
C. Pixels per inch
D. Frames per second

12. Path animation moves an object:
A. Along a predefined route
B. Without motion
C. Randomly
D. Only vertically

13. Which colour model is subtractive?
A. CMYK
B. HSV
C. HSL
D. RGB

14. Hypermedia is best described as:
A. A printed manual
B. A single video file
C. Multimedia with hyperlinks
D. Text-only media

15. Morphing changes:
A. Audio pitch
B. Font size
C. One shape into another
D. File format

16. HTML is used to:
A. Structure web content
B. Store audio samples
C. Compress video
D. Model 3D objects

17. Hertz measures:

A. Resolution
B. File size
C. Frequency
D. Loudness

18. A licence gives:

A. Ownership of all works
B. Unlimited bandwidth
C. Permission to use a work
D. Free hardware

19. Resolution refers to:

A. File type
B. Number of pixels in an image
C. Colour name
D. Font size

20. A graphics tablet is used for:

A. Printing
B. Drawing input
C. Sound output
D. Web browsing

21. Anti-aliasing is used to:

A. Smooth jagged edges
B. Increase file size only
C. Add sound
D. Compress video

22. An authoring tool is used to:

A. Assemble multimedia elements
B. Only print text
C. Only record audio
D. Only browse the web

23. Card-based authoring tools are suited to:

A. Messages with a beginning and end
B. Audio editing only
C. 3D modelling only
D. Live video only

24. Which format is commonly used for bitmap images?

A. MIDI
B. PNG
C. HTML
D. MPEG

25. Convergence in multimedia refers to:

A. Printing images
B. Deleting media files
C. Merging of media and computing
D. Recording audio only

26. Virtual reality creates:

A. A text font
B. A camera lens
C. A printed page
D. A simulated environment

27. A storyboard is:

A. A sound file
B. A video codec
C. A visual sequence plan
D. A colour model

28. A prototype is:

A. A font
B. An early model of a product
C. A final release
D. A codec

29. Which is an example of linear multimedia?

A. A cinema presentation
B. A website with menus
C. An interactive kiosk
D. A game with choices

30. Amplitude of a sound wave relates to:

A. Loudness
B. Pitch
C. Frame rate
D. Colour

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PART D (40 MARKS)

QUESTION 1

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

QUESTION 2

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(3 marks)

b) Describe TWO roles of hyperlinks in the website.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(4 marks)

QUESTION 3

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(2 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(4 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

QUESTION 4

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

(2 marks)

b) Describe TWO roles of hypertext in multimedia.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

(5 marks)

END OF QUESTION PAPER